

VitalsPatch

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Introduction

VitalsPatch is a wearable health monitoring system designed for clinical and care environments where a single practitioner must keep watch over many patients at once. The device continuously measures heart rate, blood oxygen saturation (SpO2), body temperature, and motion, and detects falls and out-of-range vital signs in real time.

Sensor data is collected on a wearable module, processed on an STM32F4 microcontroller, and forwarded to a Raspberry Pi 4 that hosts a centralized dashboard displaying live trends and prioritized alerts for multiple patients simultaneously.

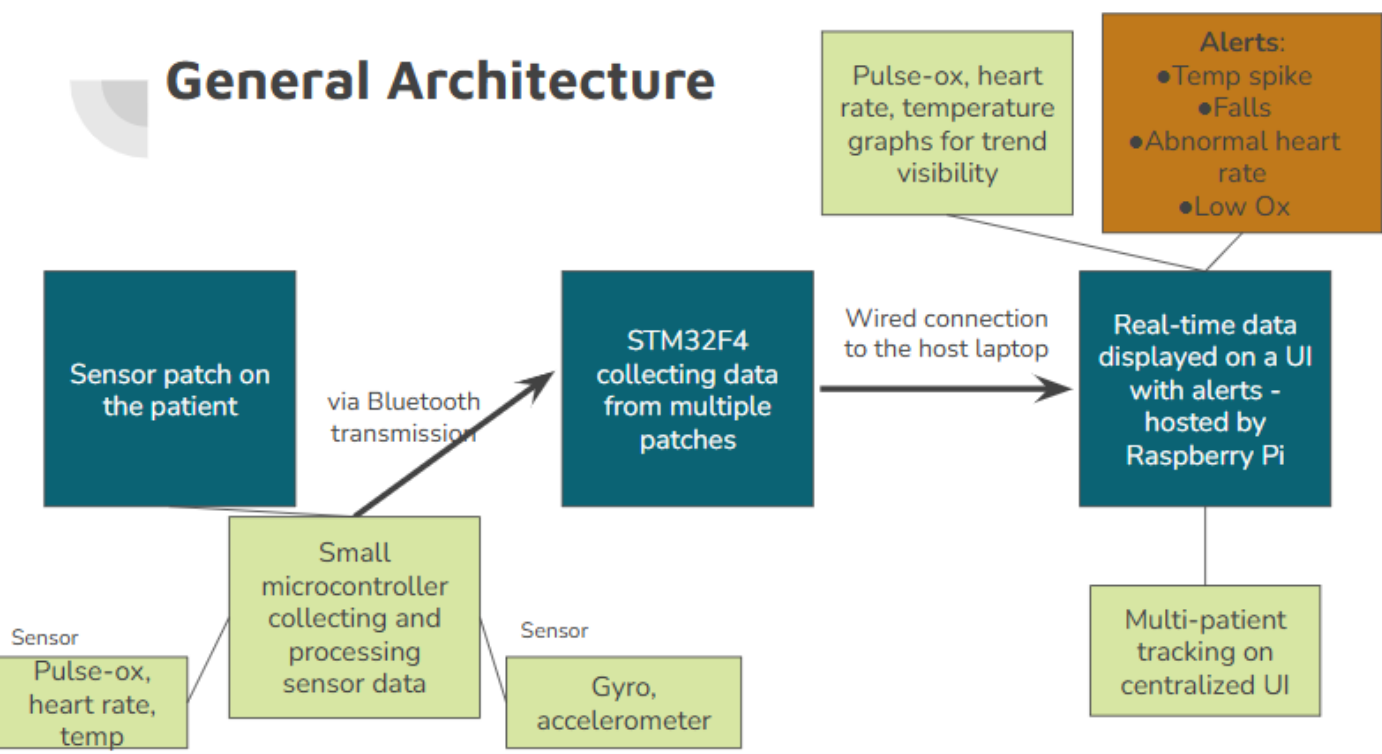
VitalsPatch is designed to improve care in a variety of healthcare settings by allowing for real-time responses and simultaneous monitoring

VitalsPatch Quick Facts

- > **Monitors key vitals:**
 - > Heart Rate
 - > Temperature
 - > Pulse Ox
 - > Abnormal movements
- > **Multi patient tracking on a centralized dashboard**
- > **Real-time alerts for abnormal readings and vitals to allow for prompt care**
- > **Uses Bluetooth connection to a localized server, allowing for remote monitoring.**

Objectives

- VitalsPatch is designed to help healthcare professionals and care staff that oversee a wide range of patients provide real-time, informed care.
- > Facilities where patients are not continuously connected to a monitor
 - > Nursing Homes
 - > Outpatient Clinics
 - > At Home Care for the Elderly and Disabled
 - > Prompt response to incidents improves outcomes
 - > Care staff can easily monitor multiple patients simultaneously



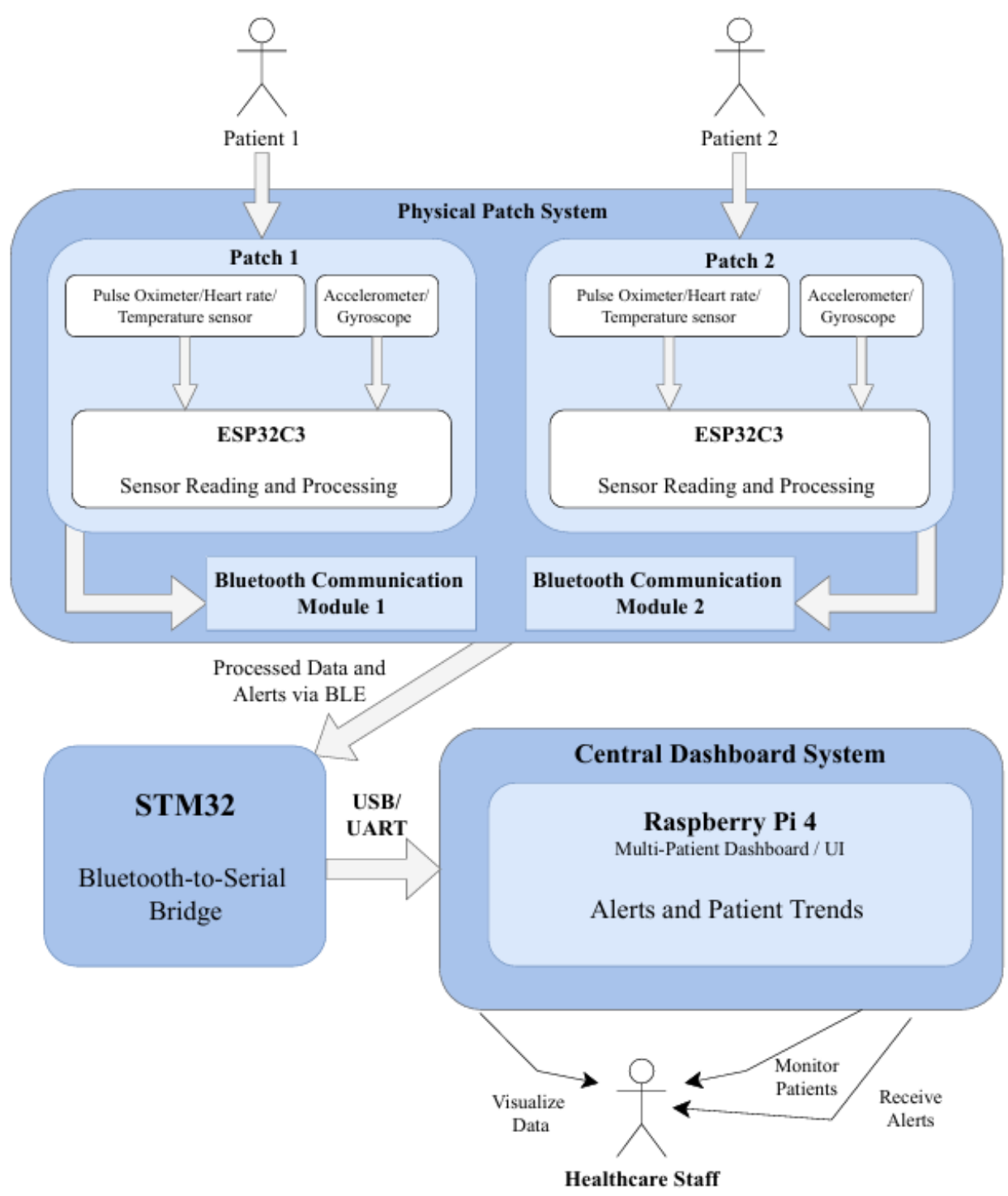
This diagram demonstrates how the entire product works from patient data to UI displays

Results

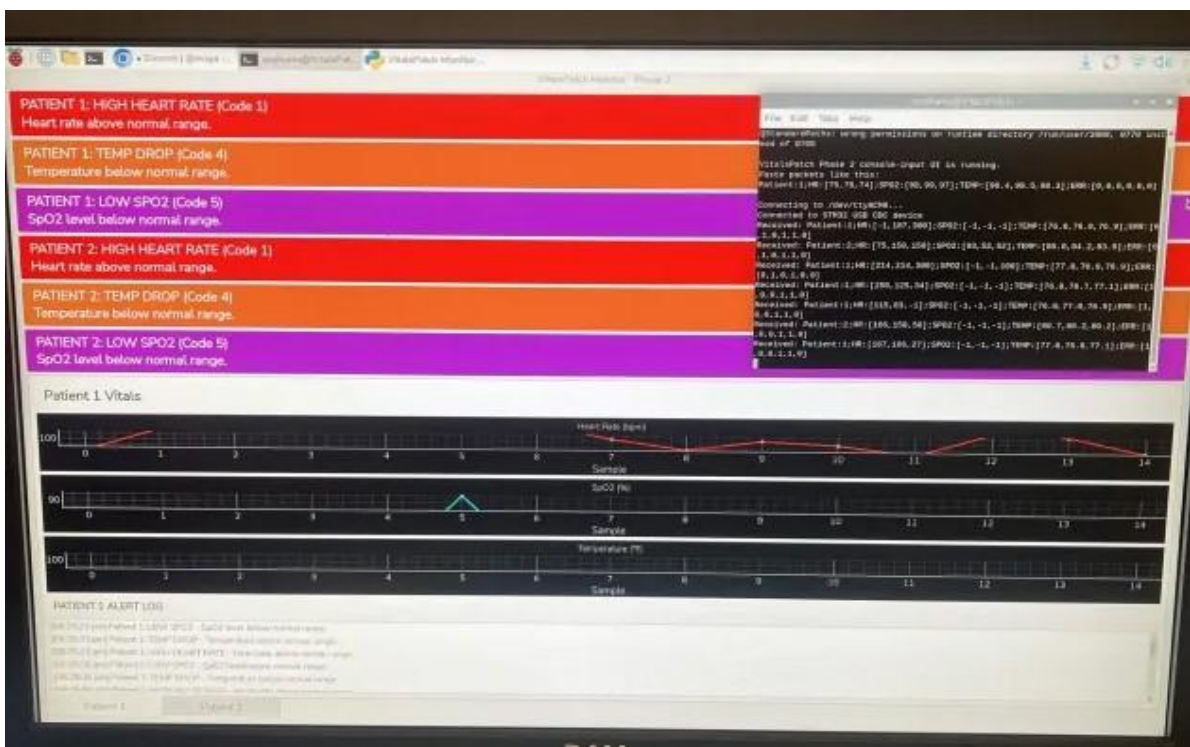
- VitalsPatch displays graphical tracking of each vital for all patients and posts alerts for all patches using a banner at the top of the interface
- > Regular heart rate is measured between 60 and 120 bpm
 - > A SPO2 above 90 was considered normal
 - > Normal temperature ranged between 95 and 100.4 Fahrenheit (Skin temperature is lower)
 - > Fall detection checked for rapid acceleration in the downward direction and then a complete stop in acceleration (impact)
 - > An alert log is displayed at the bottom, allowing providers to see past incidents

Methods

- We developed a pipeline that utilized these components:
- > Two off-the-shelf sensors collect data and transmit to an ESP32C3 on a wired connection. A combined HR, SPO2, and temperature sensor and a combined accelerometer and gyroscope
 - > ESP32C3 analyzed the data received from the two sensors and transmitted via BLE to an HM19 module
 - > An HM19 module was connected via a UART connection to the STM32
 - > Each patient patch has its own set of sensors, ESP32C3 and HM19 module
 - > The STM32 acted as a bridge between the wearables and the user interface to allow for the most efficient scalability
 - > Raspberry Pi 4 hosted the user interface that displayed alerts and vital graphs for each patch. Received data from the STM32 using a USB connection.



UML chart showing the use cases and design flow of our device

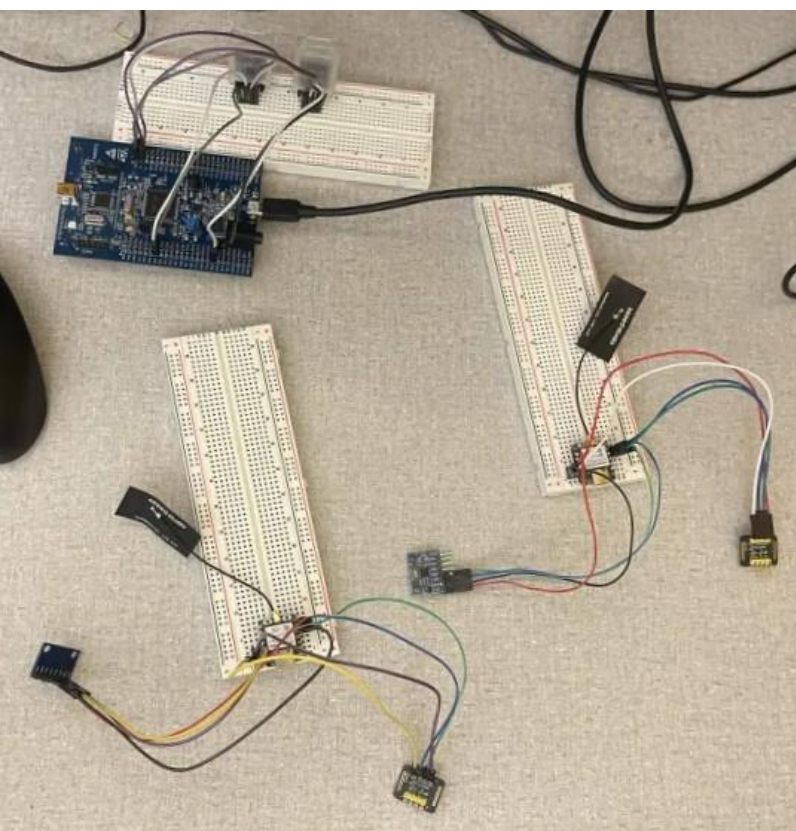


The user interface for two patients showing alert codes for both as well as the data over time for patient 1.

Conclusion

VitalsPatch demonstrates the feasibility and utility of a compact, low-cost wearable that can provide continuous monitoring in healthcare settings. By allowing for tracking of multiple wearers and providing automated real time alerts, VitalsPatch allows for more prompt care and eases the burden on staff by eliminating the need for repeated manual checks and monitoring many separate devices.

VitalsPatch utilizes individual calibration to each user in order to collect accurate data and present individually relevant alerts. This design bridges a common market gap on similar devices that are universally calibrated to paler skin tones, leading to less accuracy for darker tones, potentially impacting the quality of care.



Set up illustrating the sensors, ESP32, HM19 Bluetooth connections, and STM32, showcasing the general pipeline